

llmXive Automated Review of AgenticSTS: A Bounded-Memory Testbed for Long-Horizon LLM Agents

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: [qwen.qwen3.5-122b](#)

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper presents a well-structured argument for a bounded-memory contract, and the internal consistency of the reported results (e.g., the 3/10 vs 6/10 win rates and the corresponding statistical tests) is generally sound. The authors are transparent about the directional nature of their findings and the limitations of their sample size.

However, there are significant factual claims regarding the specific models used as baselines that cannot be verified against the public record. The paper repeatedly cites “Gemini 3.1 Pro” and “DeepSeek V4 Pro” (Sections 1, 5, 6, and Table 2) as the specific backbones evaluated. As of the paper’s publication date (May 2026), these specific model versions do not exist in the public domain or official release logs of the respective providers. This constitutes a potential hallucination of model names or a misattribution of the actual models used (e.g., perhaps “Gemini 1.5 Pro” or “DeepSeek V3”). Since the core comparison relies on these specific backbones to demonstrate the “backbone-sensitive” nature of the skill stack, the inability to verify the existence of these models undermines the reproducibility and factual accuracy of the comparative results.

Additionally, while the statistical pooling in Section 6.2 is mathematically consistent with the table data, the presentation of the pooled p-value (0.148) as a primary finding for the “scaffolded vs. unscaffolded” comparison requires a slight qualification to distinguish it from the pre-registered per-cell analysis, ensuring the reader understands the statistical weight of the aggregated data versus the individual ablation cells.

The citations to community databases (Spire Codex, STS2 Fun) are appropriate for context, but the specific “May 2026” snapshot dates should be cross-referenced with the bibliography’s access dates to ensure no temporal drift in the data sources.

Action items.

- **[writing]** The paper presents a well-structured argument for a bounded-memory contract, and the internal consistency of the reported results (e.g., the 3/10 vs 6/10 win rates and the corresponding statistical tests) is generally sound. The authors are transparent about the directional nature of their findings and the limitations of their sample size. However, there are significant factual claims regarding the specific models used as baselines that cannot be verified against the public record. The paper rep

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

The figure provides a clear visual overview of the proposed memory contract and testbed. However, the caption contains a broken cross-reference and fails to define the specific ‘L1-L5’ layers shown in the diagram.

Figure 2

Figure 2 effectively visualizes the proposed architecture and comparative scores, but it presents illustrative summary labels as definitive facts and compares distributions across significantly different sample sizes without sufficient visual or textual caveats in the panel itself.

Figure 3

The figure effectively visualizes the token audit and counterfactual growth, but the legend notation in panel (a) is slightly ambiguous regarding the formula representation, and the x-axis definition could be more explicit in the context of the caption’s description of run mechanics.

Figure 4

The figure effectively visualizes the ablation surface configuration states, but the caption’s reference to ‘adjacent bars’ contradicts the flowchart format, and the ‘L5 source’ swap is not explicitly labeled in the diagram.

Figure 5

The figure presents ascension data but fails to define the ‘A#’ labels on the x-axis or the orange/green bar colors in the legend. Additionally, the omission of sample sizes for the final two conditions makes it impossible to evaluate the robustness of those results.

Figure 6

The figure effectively visualizes the competitor comparison across effect, speed, and cost, but the caption contains a significant truncation error at the end, and the cost axis labels could be clearer.

Figure 7

The figure effectively visualizes the context growth difference between bounded and unbounded agents, but the x-axis comparison is misleading given the caption’s disclaimer that run lengths are not comparable. Additionally, the legend and y-axis label do not fully align with the caption’s definition of the AgenticSTS metric (partial vs. full prompt).

Figure 8

The figure effectively visualizes the cost-effect frontier, but the legend uses specific method names not defined in the caption, and the x-axis label fails to distinguish between estimated and measured token counts.

Figure 9

The figure fails to match its caption; it shows a single combat screenshot instead of the promised side-by-side comparison of combat and shop planning states, and it displays a game UI rather than the prompt data described.

Action items.

- **[writing]** Figure 1: The caption contains a broken cross-reference (‘are given in –’) instead of citing the specific figure or section containing the exact numbers and caveats.
- **[writing]** Figure 1: The ‘L1 protocol’ through ‘L5 skills’ labels in the bottom-left panel are not defined in the caption, leaving the specific meaning of these layers ambiguous to the reader.
- **[writing]** Figure 2: The caption states that summary labels (e.g., ‘no trade-off’, token-efficiency, score-gap figures) are illustrative, but the figure itself displays these specific claims as definitive facts (e.g., ‘90%+ fewer tokens’, ‘+52.4 vs best public’) without visual distinction, potentially misleading readers about the data’s status.
- **[science]** Figure 2: Panel (c) presents a box plot for ‘Ours +L5 scaffold’ with $n=30$, while other groups (AGI-Eval GPT-5.4, Opus 4.7) have $n=9$; the caption notes sample size differences but does not explicitly warn that the visual comparison of distributions across these disparate sample sizes is statistically limited.
- **[writing]** Figure 3: The legend in panel (a) displays the formula as ‘ $1/4 O(c^2)$ ’, but the caption describes the counterfactual as ‘ $1/4$ of naive $O(c^2)$ growth’ with a specific median token formula; the legend notation is ambiguous and should explicitly state it represents the total cumulative growth curve to match the y-axis.
- **[writing]** Figure 3: The x-axis label ‘strategic decisions c ’ is present, but the caption refers to ‘ten fixed-A0 runs’ and ‘two per cell’ without explicitly defining what the specific x-axis values (0 to 140) represent in the context of the ‘Per-run growth’ panel (e.g., is it the max decisions reached in a run?).
- **[science]** Figure 4: The diagram labels the transition from ‘mode-a’ to ‘mode-b’ as ‘swap L5 source’, but the visual content shows ‘L5 human’ (mode-a) changing to ‘L5 agent’ (mode-b). The caption defines the comparison as isolating ‘L5 source’, yet the diagram does not explicitly label ‘human’ and ‘agent’ as the specific sources being swapped, creating ambiguity about the ablation axis.
- **[writing]** Figure 4: The caption states ‘adjacent bars are organized by a named ablation axis’, but the figure is a schematic flowchart of configuration states, not a bar chart. This

terminology mismatch between the caption and the visual representation is confusing.

- **[science]** Figure 5: The x-axis labels ‘A2’, ‘A3’, ‘A4’, ‘A5’, ‘A7’, ‘A8’ are undefined; the caption does not explain what these codes represent or how they relate to the named conditions (e.g., ‘baseline-strict’, ‘Mode B’). - severity: science text:
- **[science]** Figure 5: The ‘Mode B’ and ‘full+postrun’ groups lack sample size (n) labels, whereas all other groups explicitly state ‘(n=...)’, preventing assessment of statistical reliability for these key conditions. summary:
- **[writing]** Figure 6: The caption ends abruptly with ‘e [fig_cmp_strips.pdf]’, cutting off the explanation of the token convention (likely ‘estimated’) and leaving the sentence incomplete.
- **[science]** Figure 6: The y-axis for (c) Cost is labeled ‘Fresh tokens / score pt’ with a log scale, but the data labels (e.g., ‘570.7k’) are placed directly on the plot without a clear visual mapping to the axis ticks, making it difficult to verify the values against the scale.
- **[science]** Figure 7: The x-axis extends to 1000+ decisions, but the caption states the AgenticSTS runs make ‘~100 strategic calls’ and the x-extent is ‘not comparable’. Plotting the bounded contract (dashed line) on the same x-axis as the unbounded competitors implies a direct comparison of growth over the same run length, which the caption explicitly denies, creating a misleading visual comparison.
- **[writing]** Figure 7: The legend entry ‘AgenticSTS bounded contract (~5k/call, est.)’ is ambiguous regarding the y-axis unit. The caption clarifies this is ‘strategic user-message median’ excluding the ‘constant cached system prefix’, but the y-axis label is ‘Prompt tokens per LLM call’. This discrepancy (partial vs. full prompt) is not visually distinguished in the legend or axis, potentially confusing the reader about what is being measured.
- **[writing]** Figure 8: The legend labels ‘AgenticSTS (full-frozen)’ and ‘AgenticSTS (baseline-strict)’ are not defined in the figure caption, which only refers to them generically as ‘our cells’.
- **[science]** Figure 8: The x-axis label ‘Fresh LLM tokens per run’ is ambiguous because the legend explicitly states that the diamond markers (AgenticSTS) use ‘estimated’ tokens based on a convention, whereas the competitor markers use ‘measured’ tokens; the axis label should reflect this mixed data source.
- **[science]** Figure 9: The caption claims the figure shows ‘combat planning (left) and shop planning (right)’, but the rendered image is a single screenshot of a combat encounter with no right-side panel or shop interface visible.
- **[science]** Figure 9: The image displays a game UI (Slay the Spire) rather than the ‘decision states used in the prompt’ (e.g., text logs or structured data) described in the caption.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The paper is generally well-structured and avoids the worst excesses of in-group shorthand, successfully defining its core “bounded contract” concept early on. However, a competent reader from an adjacent field (e.g., a researcher in standard RL or NLP) would likely stumble on a few specific instances of undefined notation and in-group naming conventions.

First, in Section 4.1 (Equation 1), the symbol $\circ$ is introduced as the composition operator. In the broader ML literature, π almost exclusively denotes a policy function. Here, it is repurposed as a prompt assembly function. While the context makes the intent clear, the lack of an explicit definition (“where $\circ$ is the composition function...”) forces the reader to infer the meaning, which is a minor barrier.

Second, the paper relies heavily on specific, code-like labels for its experimental conditions and architectural layers (e.g., “mode-a”, “mode-b-frozen”, “operator prompts”, “state-typed prompts”). While these are defined in the text, they are presented as proper nouns without immediate operational glosses. For a reader outside this specific subfield, distinguishing “mode-a” from “mode-b” requires careful back-and-forth reading. A brief parenthetical explanation at the first mention of each (e.g., “mode-a (human-authored L5 seed bodies)”) would significantly improve flow and accessibility.

Finally, the term “MCP” appears in Section 7 (referencing STS2MCP and MCP servers) without expansion. Given that “Model Context Protocol” is a specific, emerging standard in the agentic tooling space but not yet universal knowledge across all of AI, it should be spelled out at first use to ensure the reader understands the technical dependency.

These are minor friction points that can be resolved with simple parenthetical expansions, but they currently represent the primary jargon-related barriers to a smooth read for an adjacent-field expert.

Action items.

- **[writing]** Section 4.1, Eq. 1: The symbol $\circ$ is used as the composition operator without definition. While π is standard for policies, here it denotes a prompt assembly function. Define it explicitly: ‘where $\circ$ is the prompt composition function that concatenates...’
- **[writing]** Section 4.2: The terms ‘operator prompts’ (L1), ‘state-typed prompts’ (L2), and ‘game knowledge’ (L3) are introduced as proper nouns for specific layers. While the context implies their function, a brief parenthetical gloss at first use (e.g., ‘L1 operator prompts (immutable role templates)’) would aid an adjacent-field reader.
- **[writing]** Section 4.3: The routing tiers ‘fast’, ‘strategic’, ‘analysis’, and ‘evolution’ are used as specific technical categories. Define the criteria for each tier (e.g., ‘fast: trivial combat plans; strategic: ordinary decisions’) in the first sentence of the paragraph to clarify the distinction for non-specialists.

- **[writing]** Section 5.1: The condition names ‘mode-a’, ‘mode-b-frozen’, and ‘full-frozen’ are used as specific experimental labels. While defined in the text, they appear as code-like identifiers. Ensure the first mention of each includes a brief operational description (e.g., ‘mode-a (human-authored L5 seed bodies)’ to prevent confusion with generic modes.
- **[writing]** Section 7: The term ‘MCP’ is used in ‘STS2MCP’ and ‘MCP server’ without expansion. While likely ‘Model Context Protocol’ in this ecosystem, it is not a universal standard. Define the acronym at first use in Section 7 or the Introduction.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The paper’s argument structure is logically sound and internally consistent. The central thesis—that a bounded, typed memory contract enables ablation and inspection of long-horizon agent decisions—is supported by a coherent chain of premises and conclusions.

1. **Premise-Conclusion Alignment:** The introduction establishes the problem (unbounded context growth obscures causal attribution) and proposes the solution (typed retrieval). The results section (Section 6) directly addresses this by presenting a fixed- A_0 ablation matrix where the “no-scaffold” baseline is compared against configurations with specific layers (L_4 , L_5) enabled. The conclusion that L_5 (skills) drives the largest observed lift ($3/10 \rightarrow 6/10$) follows directly from the data in Table 1, and the authors correctly qualify this as “directional” rather than statistically significant ($p \approx 0.37$), avoiding the non-sequitur of claiming definitive causality from a small sample.
2. **Consistency of Definitions and Numbers:** The five-layer architecture (L_1 – L_5) is defined in Section 4 and used consistently throughout the methodology (Section 5) and results (Section 6). The sample sizes are consistent: the headline fixed- A_0 analysis uses a balanced subset of 50 games (5×10), while the total archive contains 298 trajectories. The text explicitly distinguishes between the headline ablation and the diagnostic streams (cross-backbone, ladder), preventing scope inflation. The derived score formula (Eq. 1) is applied consistently in the text and the appendix tables.
3. **Handling of Limitations:** The paper avoids internal contradiction by explicitly stating in the Limitations (Section 10) and Conclusion (Section 9) that the comparison to accumulating-context agents is “operational” rather than a controlled ablation. This aligns with the methodology, which notes that a matched same-codebase accumulating-context cell is future work. The claim that the bounded contract “outperforms” competitors is carefully framed as an observation of the current state of practice (Section 7) rather than a proven causal superiority of the contract itself, which is logically consistent with the lack of a controlled

baseline.

4. **No Circular Reasoning:** The argument does not assume the conclusion. The effectiveness of the L_5 layer is an empirical finding derived from the ablation, not a premise used to justify the architecture. The “bounded” nature of the contract is a design choice, and its benefits (inspectability, ablation) are demonstrated, not assumed.

The reasoning holds together: the premises (design of the testbed, ablation results) support the conclusions (typed retrieval isolates variables, L_5 is the primary driver of performance in this setup) without logical gaps or contradictions.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper demonstrates exceptional discipline in aligning its rhetorical scope with the actual evidentiary boundaries of its experiments. Unlike many works in the agentic systems space that extrapolate single-benchmark wins to universal solutions, this manuscript consistently frames its contributions as specific to the “bounded-memory contract” within the *Slay the Spire 2* testbed.

The abstract explicitly qualifies the headline result (3/10 vs 6/10 wins) as “directional rather than statistically decisive” ($p \approx 0.37$), preventing the common overreach of presenting a small-sample trend as a definitive breakthrough. Similarly, the introduction and results sections carefully distinguish between “operational comparisons” (e.g., against STS2MCP and CharTyr) and “controlled tests,” acknowledging that differences in game patches and routing prevent a clean causal attribution of the performance gap solely to the memory contract.

The “Limitations” section is robust and substantive, directly addressing the most significant scope boundaries: the lack of a same-codebase accumulating-context baseline, the restriction to a single character (Silent), and the training-free nature of the evaluation. The conclusion mirrors the body’s caution, stating that whether a bounded contract “outperforms a matched accumulating-context design remains an open question,” rather than asserting it as a proven fact.

There are no instances of “solves,” “proves,” or “universally” language that exceed the data. The paper successfully avoids the trap of presenting a novel testbed as a solved problem, instead positioning it as a reusable methodology for future inquiry. The scope of the claims is tightly bound to the 298 trajectories and the specific ablation matrix presented.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

This paper presents a low-risk research artifact: a bounded-memory testbed for LLM agents operating within a single-player video game environment (Slay the Spire 2). The work does not involve human subjects, personal data, or sensitive real-world interactions.

The “Ethical Considerations” section (sec/11.ethics.tex) is present and appropriately scoped. It correctly identifies that no human subjects are involved, no personal data is collected, and the game/mod licensing is respected (using local regeneration of game strings rather than redistributing copyrighted assets). The authors explicitly address the “Risk of dual use,” noting that while the underlying agent capabilities are general, the current deployment is confined to a game setting with low direct harm, and they acknowledge that deployment in real-world domains would require separate safety reviews.

There are no indications of:

- Dual-use capabilities described without mitigation (the method is a memory interface for a game agent).
- Human-subjects data without consent (none used).
- PII or re-identifiable data exposure (trajectories are game states).
- Scraped data used against license terms (the paper explicitly avoids redistributing game assets and cites community mods with appropriate provenance).
- Systems designed to deceive or surveil (the agent plays a game).
- Operational details for biohazard or cyber-offense (none present).

The paper is a standard methodological contribution in the AI agent space with appropriate, albeit brief, ethical disclosures suitable for the low-risk nature of the work. No action items are required.

Scientific Evidence

Weights the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

The paper presents a novel testbed for bounded-memory agents and is remarkably transparent about the statistical limitations of its primary results. The authors correctly identify that the headline win-rate improvement (3/10 to 6/10) is “directional rather than statistically decisive” (p 0.37). However, the evidentiary strength of the paper relies heavily on interpreting these directional trends and operational comparisons as support for the “bounded contract” hypothesis.

The primary concern is the sample size relative to the claimed effect size in the fixed-A0 ablation. While the authors provide Wilson confidence intervals, the overlap between the “No

scaffold” and “Hand skills” conditions is substantial. The claim that the L5 layer is the driver of the gain is plausible but not statistically established by the current $N=10$ per cell. A skeptical reader cannot rule out that the observed 30% lift is a lucky seed or sampling noise. To strengthen the evidence, the authors should report the bootstrap confidence interval for the *difference* in win rates (or scores) rather than just the individual intervals, and ideally increase the sample size to $N=20-30$ per cell to narrow the confidence bands enough to make a definitive claim about the L5 contribution.

Secondly, the comparison with external competitors (STS2MCP, CharTyr) in Section 7 is a significant confound. The authors admit to differences in game patches, routing strategies, and prompt cadences. While they frame this as an “operational comparison,” the large performance gap (0/5 wins vs 6/10) is presented as evidence of the bounded contract’s superiority. Without a controlled experiment where the competitors run on the same game patch, with the same routing logic, and the same prompt budget, the design cannot rule out that the performance gap is due to these implementation differences rather than the memory contract itself. The paper should either run a controlled ablation of the competitors or explicitly limit the claim to “our system outperforms current open-source implementations under their default configurations” without implying the contract is the sole cause.

Finally, the cross-backbone transfer results ($N=5$ per cell) are too sparse to support claims about “backbone sensitivity.” With 0/5 wins in all conditions, the score differences are the only signal, but the variance in such a small sample is high. The evidence here is suggestive but not conclusive. Increasing the sample size or providing a more robust statistical analysis of the score distributions would be necessary to support the claim that the skill stack is backbone-sensitive.

Overall, the paper is a strong contribution to the field of agent evaluation, but the specific claims about the efficacy of the L5 layer and the superiority of the bounded contract over existing methods require more robust statistical evidence or clearer framing as preliminary/operational findings.

Action items.

- **[writing]** The paper presents a novel testbed for bounded-memory agents and is remarkably transparent about the statistical limitations of its primary results. The authors correctly identify that the headline win-rate improvement (3/10 to 6/10) is “directional rather than statistically decisive” (p 0.37). However, the evidentiary strength of the paper relies heavily on interpreting these directional trends and operational comparisons as support for the “bounded contract” hypothesis. The primary concern is

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The statistical treatment in this paper is exemplary for the field of LLM agent evaluation,

particularly given the small sample sizes inherent to long-horizon game runs. The authors correctly identify that their primary headline result (3/10 vs 6/10 wins) is underpowered for a definitive significance claim and explicitly label it “directional” rather than “statistically significant.”

Specifically, the choice of the Fisher exact test for the 2×2 contingency table (Section 6.2) is the correct parametric-free approach for small counts ($N = 10$ per cell), and the reported $p \approx 0.37$ is consistent with the observed data. The use of Wilson score intervals for win rates (Section 5.3, Appendix) is appropriate for binomial proportions, especially where p is near 0 or 1 or n is small, avoiding the under-coverage issues of the standard Wald interval. The bootstrap method (5,000 resamples) for continuous score distributions is also a robust choice given the non-normal nature of game scores.

Crucially, the authors avoid the common pitfall of “p-hacking” or over-interpreting non-significant trends. They do not claim the L_5 skill layer is “significantly better” despite the point estimate lift; instead, they transparently report the overlap in confidence intervals and the high p-value. The separation of the fixed- A_0 ablation (balanced $N = 50$) from the diagnostic cross-backbone and ladder streams prevents inappropriate pooling of heterogeneous data. The reporting of exact denominators (e.g., “0/5” for cross-backbone wins) and the explicit distinction between “operational comparisons” and “controlled tests” further ensures that the numbers mean exactly what the text claims. No statistical errors, missing uncertainty measures, or misapplied tests were found.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The paper is generally well-structured and the core argument regarding bounded memory contracts is presented with logical progression. However, several instances of dense phrasing and abrupt transitions impede the reader’s momentum, requiring re-reading to parse the intent.

In the Abstract, the final sentence is overloaded with a complex noun phrase (“an agent design and a validated, reusable methodology..”) that serves as the conclusion. This creates a “garden-path” effect where the reader must hold the entire list of releases in memory before understanding the ultimate purpose. Splitting this into two distinct sentences would significantly improve readability.

Section 3.2 contains a sentence defining the score formula that is interrupted by a parenthetical clarification about data provenance (“not copied from the raw archive score field”). This interruption breaks the logical flow of the mathematical definition. Moving this clarification to a separate sentence or integrating it earlier would allow the reader to process the formula without distraction.

In Section 4.1, the transition from theoretical complexity analysis (O notation) to the empirical

linearity audit (Figure 3) is abrupt. The text jumps directly to “Figure 3a–b reports...” without a bridging phrase indicating that the figure serves as empirical validation of the preceding theory. A simple transition like “To validate this theoretical bound, we conducted...” would smooth this handoff.

Section 6.2 introduces Equation 2 with the phrase “Writing the layer-attributable difference as...” without first explicitly defining the concept in plain English. The reader must parse the equation to understand what Δ_{L_ℓ} represents. A preceding sentence explicitly stating “We define the layer-attributable difference as the gap in win rates...” would make the equation’s introduction more intuitive.

Finally, the Discussion section (Section 7) opens with a slightly abstract sentence (“The main lesson is that...”). While grammatically correct, rephrasing to be more direct (“The primary finding is that...”) would strengthen the impact and clarity of the conclusion.

These issues are minor and do not obscure the scientific contribution, but addressing them would allow the reader to move through the text with greater ease and confidence.

Action items.

- **[writing]** The paper is generally well-structured and the core argument regarding bounded memory contracts is presented with logical progression. However, several instances of dense phrasing and abrupt transitions impede the reader’s momentum, requiring re-reading to parse the intent. In the Abstract, the final sentence is overloaded with a complex noun phrase (“an agent design and a validated, reusable methodology...”) that serves as the conclusion. This creates a “garden-path” effect where the reader mus